

Noising Depth Controls Validity and Identity Preservation in Diffusion Counterfactual Explanations for Abdominal CT

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Abstract—Counterfactual explanation identifies the smallest change to an input that alters a classifier’s decision, and diffusion models are increasingly used to generate such changes. The procedure has a free parameter that is seldom reported: the depth to which the input is noised along the forward diffusion trajectory before guided denoising begins. We train an unconditional denoising diffusion probabilistic model on 34,561 abdominal CT slices, add classifier-guided sampling, and evaluate seven noising depths on 256 held-out slices, measuring validity, proximity, pixel-level sparsity and identity preservation. Validity is non-monotonic in noising depth. It reaches 0.426 at $t_{\text{start}} = 200$ and falls to 0.219 at $t_{\text{start}} = 600$, while identity correlation with the input decreases monotonically from 0.981 to 0.079. At the largest depth the procedure modifies 88% of pixels. We define an operating criterion based on the identity statistic and the location of the validity peak, and report a nearest-neighbour memorisation audit with a real-data control, which found no evidence of training-image reproduction. Code, result files and figures are available.

Index Terms—explainable AI, counterfactual explanation, diffusion models, computed tomography, generative models, memorisation

I. INTRODUCTION

Saliency maps indicate which regions of an input a network responded to. They do not indicate what change to the input would alter the network’s decision. A counterfactual explanation addresses that question: given an image the model labels y , it produces the nearest image the model labels y' . The usefulness of the explanation depends on the location and extent of the edit.

Diffusion models define a continuous path between data and noise, which makes them suitable for generating such edits. The standard procedure noises a real image forward to timestep t_{start} and denoises it under gradient guidance from the classifier being explained [1], [2]. The choice of t_{start} determines a trade-off between the probability of changing the prediction and the amount of the original image that is preserved.

This paper reports a systematic study of that parameter. Our contributions are:

- 1) A sweep of noising depth on abdominal CT, measuring validity, proximity, sparsity and identity preservation on a common set of 256 slices, which shows that validity is non-monotonic in noising depth.
- 2) A quantitative criterion for selecting an operating point, based on the identity statistic and the location of the validity peak.
- 3) A nearest-neighbour memorisation audit with a real-data control, following the finding that latent diffusion models can reproduce training patients [5].

II. RELATED WORK

Counterfactual explanation. DiME [1] introduced diffusion-based counterfactual generation for natural images. In medical imaging, diffusion autoencoders have been used for classification and regression explanations [2], and counterfactual diffusion has been applied to histopathology. Salahuddin et al. [3] combine counterfactual explanation with uncertainty estimation for renal cyst segmentation on contrast-enhanced abdominal CT using a VAE-GAN, and validate across centres. The present work substitutes a diffusion model for the VAE-GAN and studies the sampling parameter that substitution introduces.

Interpretability evaluation. Salahuddin et al. [4] survey interpretability for medical image analysis and note that explanation quality is frequently asserted without measurement. We report four quantitative axes.

Memorisation. Dar et al. [5] show that unconditional latent diffusion models trained on medical images can reproduce training patients, with direct implications for sharing synthetic data. We adopt their framing and report a nearest-neighbour statistic against a real-data control.

III. METHOD

A. Data

We use OrganAMNIST from MedMNIST v2 [6], the 64×64 release. The dataset contains axial abdominal CT slices cropped to eleven organ classes and derived from the Liver Tumor Segmentation Benchmark. Splits are 34,561

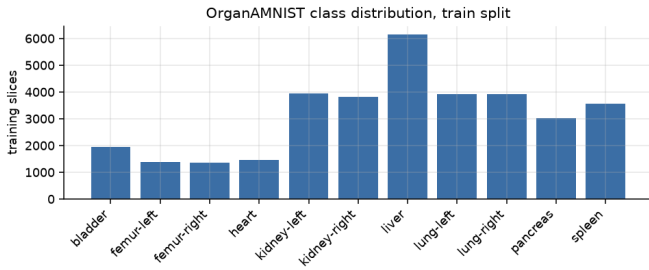


Fig. 1. OrganAMNIST training class distribution. The imbalance motivates per-class reporting in Table I.

training, 6,491 validation and 17,778 test images. Intensities are mapped from uint8 to $[-1, 1]$. Fig. 1 shows the class distribution, which is uneven: liver contributes 6,164 training slices and femur-left 1,390. Per-class classifier accuracy is reported in Table I.

B. Generative model

The diffusion model is a MONAI DiffusionModelUNet with 10.4M parameters, channel widths (64,128,192), two residual blocks per level and self-attention at the coarsest resolution only. It is trained to predict the noise ϵ added at timestep t under a DDPM schedule with $T = 1000$ [7], minimising

$$\mathcal{L} = \mathbb{E}_{x_0, \epsilon, t} \|\epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, t)\|_2^2. \quad (1)$$

Training ran for 40 epochs at batch size 512 in bfloat16, taking 23 minutes on one NVIDIA A100.

C. Classifier under explanation

The model being explained is a four-stage convolutional network over 64×64 slices with 11 outputs. A small classifier keeps the counterfactual search inexpensive and does not affect the quantities measured here.

D. Counterfactual generation

Given a slice x_0 with source label y and a target label $y' \neq y$, we noise x_0 forward to t_{start} and denoise with DDIM [8] over 100 steps. At each step we form the predicted clean image

$$\hat{x}_0^{(t)} = \frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}}, \quad (2)$$

and modify the noise prediction using the classifier gradient,

$$\tilde{\epsilon} = \epsilon_\theta(x_t, t) - s\sqrt{1 - \bar{\alpha}_t} \nabla_{\hat{x}_0} \log p_\phi(y' | \hat{x}_0^{(t)}), \quad (3)$$

with guidance scale $s = 6$.

The gradient is taken with respect to $\hat{x}_0^{(t)}$ because the classifier was trained on clean slices and has not been exposed to partially noised latents.

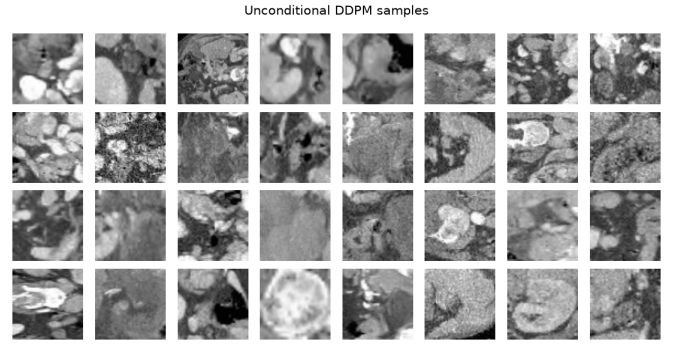


Fig. 2. Unconditional samples after 40 epochs. Organ boundaries, vessels, bright contrast-enhanced structures and dark air appear in plausible arrangements.

TABLE I
CLASSIFIER TEST ACCURACY PER ORGAN

Organ	Acc.	Organ	Acc.
heart	0.885	pancreas	0.976
kidney-left	0.899	spleen	0.982
femur-left	0.946	liver	0.992
kidney-right	0.946	lung-right	0.999
bladder	0.952	lung-left	1.000
femur-right	0.968	overall	0.958

E. Evaluation

We report four axes on 256 randomly chosen test slices, on which the classifier achieves 0.934 accuracy. **Validity** is the fraction for which the classifier predicts y' after editing. **Proximity** is mean L_1 change and the fraction of pixels moving more than 0.1 on the $[-1, 1]$ scale. **Identity** is the centred cosine similarity between x_0 and the counterfactual. **Realism** is a Frechet distance computed on features of the organ classifier instead of ImageNet Inception, since Inception features are trained on natural photographs. The value is comparable within this study only.

For memorisation we compute, for each of 1,024 generated samples, the minimum L_2 pixel distance to any of 2,048 training images, and compare the distribution against the same statistic computed for real held-out test images.

IV. RESULTS

A. Sample quality

Fig. 2 shows unconditional samples. Noise-prediction MSE fell from 0.096 to 0.034 over training. The Frechet distance between generated samples and training data is 46.8 in classifier feature space, against a floor of 2.9 measured between real test images and training data. Samples remain distinguishable from real data, which is expected at this training budget.

B. Classifier

The classifier reaches 95.8% test accuracy. Per-class results (Table I) span 0.885 for heart to 1.000 for left lung; lungs are near-perfect because they present air against soft tissue.

TABLE II
COUNTERFACTUAL QUALITY AGAINST NOISING DEPTH, 256 TEST SLICES

t_{start}	Validity	L_1	Pixels chg.	Identity
50	0.086	0.054	0.162	0.981
100	0.301	0.090	0.344	0.950
150	0.387	0.114	0.424	0.923
200	0.426	0.155	0.546	0.866
300	0.375	0.305	0.777	0.602
400	0.398	0.422	0.851	0.335
600	0.219	0.481	0.876	0.079

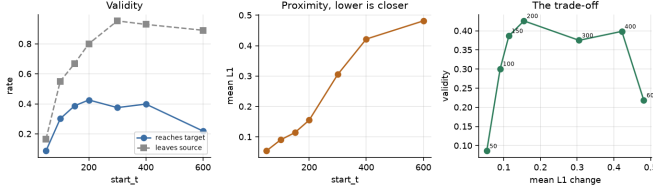


Fig. 3. Validity, proximity, and the trade-off between them. Validity peaks at $t_{\text{start}} = 200$ and declines thereafter while the L_1 change continues to grow.

C. Validity is non-monotonic in noising depth

Table II and Fig. 3 give the main result.

Validity rises steeply to 0.426 at $t_{\text{start}} = 200$, then falls to 0.219 at $t_{\text{start}} = 600$. Meanwhile L_1 change grows monotonically from 0.054 to 0.481 and identity correlation decays from 0.981 to 0.079.

The common assumption that deeper noising buys more successful flips is therefore false in this setting. Past the peak, guided denoising has enough freedom to leave the neighbourhood of the original slice entirely. The classifier is then being asked about a different image, not about a modified version of the input, and it is no longer obliged to move toward the target class. At $t_{\text{start}} = 600$ the procedure changes 88% of pixels and retains essentially nothing of the input.

We suggest a simple operating criterion: choose the largest t_{start} for which identity correlation remains above a threshold (here 0.85), and verify that validity has not yet begun to fall. Both conditions select $t_{\text{start}} = 200$.

Fig. 4 shows qualitative results at the selected operating point.

D. Memorisation audit

Generated samples sit approximately 10% closer to the training set than real unseen images (Table III), and the closest generated sample is *further* from training data than the closest real test image. There is no evidence of training-image reproduction at this scale. The audit takes minutes to run on a trained model.

V. LIMITATIONS

Results are on 64×64 single slices, not full CT volumes. The reported numbers apply at this resolution. Peak validity is 0.426, so most counterfactuals do not reach the target class; some target classes are unreachable within a fixed field of view, and a class-conditional diffusion model would likely improve

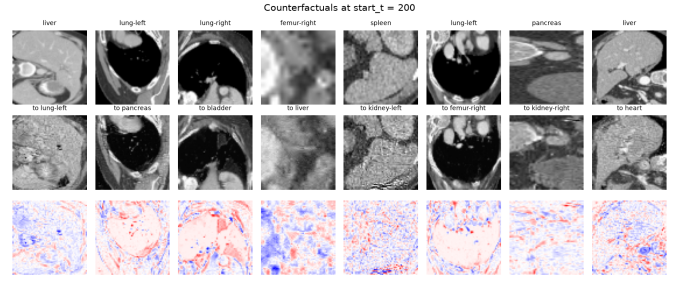


Fig. 4. Counterfactuals at the operating point. Top: original. Middle: counterfactual. Bottom: difference. Anatomy is preserved and edits are localised and structured.

TABLE III
NEAREST-NEIGHBOUR DISTANCE TO THE TRAINING SET

Query set	Median	Minimum	Ratio
Generated ($n=1024$)	22.68	11.48	0.899
Real test ($n=310$)	25.22	10.71	1.000

the remainder. Frechet distance is measured in classifier feature space and is not comparable to published FID values. Organ identity is a proxy for a clinical label; applying the pipeline to a diagnostic target such as malignancy is the natural next step. Finally, the sweep was run with a single seed, so the shape of the curve is reliable but individual values carry sampling noise.

VI. CONCLUSION

Validity is non-monotonic in noising depth, peaking at $t_{\text{start}} = 200$ and declining thereafter while identity preservation falls monotonically. Beyond the peak the procedure modifies most of the image and the output no longer corresponds to the input slice. Reporting the full curve, and the identity statistic alongside validity, allows an operating point to be selected. A nearest-neighbour memorisation audit found no evidence of training-image reproduction.

REPRODUCIBILITY

Code, trained-model configuration, result files and all figures are available at <https://github.com/Joana-Mansa/ct-diffusion-counterfactuals>. The full pipeline runs in approximately 35 minutes on one A100 and downloads its data automatically.

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